

Практична 3

Завдання 1

```
pr_3 prg > batura_k_it-31_prg > task1.py > ...
1 print("Karina Batura, IT-31")
2
3 name = input("Enter your name: ")
4 age_i = input("Enter your age: ")
5
6 if not name:
7     print("name not entered.")
8     name = "Anonymous"
9
10 if not age_i:
11     print("age not entered!")
12 else:
13     age = int(age_i)
14
15     if age < 0:
16         category = "invalid age value"
17     elif age <= 6:
18         category = "child"
19     elif age <= 17:
20         category = "schoolchild"
21     elif age <= 64:
22         category = "adult"
23     else:
24         category = "senior"
25
26 print(f"Hello, {name}! Your age category is: {category}.")
```

● PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg> & "C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg\pr_3 prg\batura_k_it-31_prg\task1.py"

Karina Batura, IT-31
Enter your name: Karina
Enter your age: 17
Hello, Karina! Your age category is: schoolchild.

○ PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg> █

Завдання 2

```
print("Karina Batura, IT-31")
n = int(input("Enter an integer:"))
if not n:
    print("the number was not given")
else:
    number = int(n)

    if number > 0:
        print("number is positive")
    elif n < 0:
        print("number is negative")
    else:
        print("number is 0")

    if number != 0:
        if number % 2 == 0:
            print("number is even")
        else:
            print("number is odd")
        if (10 <= number <= 99) or (-99 <= number <= -10):
            print("The number is two-digit")
```

PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg> & "C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg\pr_3 prg\batura_k_it-31_prg\task2.py"

Karina Batura, IT-31
Enter an integer:-67
number is negative
number is odd
The number is two-digit

○ PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg> █

Завдання 3

```

print("Karina Batura, IT-31")

first = float(input("Enter first number: "))
operation = input("Enter operation (+, -, *, /, //, %, **): ")
second = float(input("Enter second number: "))

if operation == "+":
    result = first + second
elif operation == "-":
    result = first - second
elif operation == "*":
    result = first * second
elif operation == "/":
    if second == 0:
        result = None
        print("Error: division by zero")
    else:
        result = first / second
elif operation == "//":
    if second == 0:
        result = None
        print("Error: division by zero")
    else:
        result = first // second

```

```

        result = first // second
elif operation == "%":
    if second == 0:
        result = None
        print("Error: division by zero")
    else:
        result = first % second
elif operation == "**":
    result = first ** second
else:
    result = None
    print(f"Error: unknown operation '{operation}'")

if result is not None:
    print(f"{first} {operation} {second} = {result:.4f}")

```

```

Karina Batura, IT-31
Enter first number: 4
Enter operation (+, -, *, /, //, %, **): -
Enter second number: 5
4.0 - 5.0 = -1.0000
PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg>

```

Завдання 4

```

1 print("Karina Batura, IT-31")
2 score = int(input("Enter your expected score (0-100) "))
3 missed = int(input("Enter number of missed classes: "))
4 if score < 0 or score > 100:
5     print("Error: Score must be between 0 and 100.")
6 else:
7     if score >= 90:
8         ects = "A"
9     elif score >= 82:
10        ects = "B"
11    elif score >= 74:
12        ects = "C"
13    elif score >= 64:
14        ects = "D"
15    elif score >= 60:
16        ects = "E"
17    else:
18        ects = "F"
19
20    if score >= 60:
21        status = "passed"
22    else:
23        status = "failed"
24
25    if missed / 16 > 0.30:
26        print("not included")
27    print(f"Score: {score}, ECTS: {ects}, Status: {status}")

```

```

Karina Batura, IT-31
Enter your expected score (0-100) 40
Enter number of missed classes: 4
Score: 40, ECTS: F, Status: failed
PS C:\Users\Windows 11 Pro\OneDrive\Desktop\pr_3 prg>

```

Завдання 5

prg > batura_k_it_31 prg > task3.py > ...

```
print("Karina Batura, IT-31")

day = int(input("Enter day: "))
month = int(input("Enter month: "))
year = int(input("Enter year: "))

if year <= 0:
    print("Date is invalid: year must be positive")
elif month < 1 or month > 12:
    print("Date is invalid")
else:
    if month in (1, 3, 5, 7, 8, 10, 12):
        max_days = 31
    elif month in (4, 6, 9, 11):
        max_days = 30
    else:
        if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
            max_days = 29
        else:
            max_days = 28

    if day < 1 or day > max_days:
        print(f"Date is invalid: month {month} has only {max_days} days")
    else:
        print("Date is valid")
```

```
_3 prg/pr_3 prg/batura_k_it-3
Karina Batura, IT-31
Enter day: 23
Enter month: 3
Enter year: 2009
Date is valid
```